

EA-0020 - AQELYN Engineering Archive

AQELYN - EA-0020 Engineering Archive

IS-020 - AQELYN AI Decision Intelligence Engine

****Archive ID:**** EA-0020

****Implementation Specification:**** IS-020

****Component:**** AQELYN AI Decision Intelligence Engine

****Project:**** AQELYN

****System Type:**** Cyber Security Operating Environment

****Status:**** COMPLETE

****Repository Impact:**** No top-level repository structure changes

****Breaking Changes:**** None

****Engineering Phase:**** Phase 3

****Predecessor Archives:**** EA-0001 through EA-0019

****Next Specification:**** IS-021 - AQELYN Predictive Analytics & Forecasting Engine

Document Control

| Field | Value |

---|---

| Document | Engineering Archive EA-0020 |

| Specification | IS-020 - AQELYN AI Decision Intelligence Engine |

| Publication Format | Markdown, PDF, HTML, ZIP |

| Source of Truth | MD/EA-0020.md |

| Archive Rule | Implementation Specification -> Engineering Archive -> Continue |

| Repository Rule | Fixed repository structure; no redesign |

| Completion State | IS-020 complete; EA-0020 generated |

1. Engineering Context

AQELYN is being built as a modular Cyber Security Operating Environment.

Completed engineering archive chain:

| Engineering Archive | Implementation Specification | Component |

---|---

| EA-0001 | IS-001 | AQELYN Kernel |

| EA-0002 | IS-002 | Universal Object Model |

| EA-0003 | IS-003 | AQELYN Event Bus |

EA-0004	IS-004	AQELYN Evidence Engine
EA-0005	IS-005	AQELYN Knowledge Graph
EA-0006	IS-006	AQELYN Trust Engine
EA-0007	IS-007	AQELYN Mission Engine
EA-0008	IS-008	AQELYN Workflow Engine
EA-0009	IS-009	AQELYN Policy Engine
EA-0010	IS-010	AQELYN Compliance & Governance Engine
EA-0011	IS-011	AQELYN Identity & Access Governance Engine
EA-0012	IS-012	AQELYN Asset & Configuration Governance Engine
EA-0013	IS-013	AQELYN Risk Intelligence Engine
EA-0014	IS-014	AQELYN Threat Intelligence Fusion Engine
EA-0015	IS-015	AQELYN Security Operations (SOC) Engine
EA-0016	IS-016	AQELYN Digital Forensics Engine
EA-0017	IS-017	AQELYN Threat Detection & Analytics Engine
EA-0018	IS-018	AQELYN Automated Response & Orchestration Engine
EA-0019	IS-019	AQELYN Security Data Lake & Telemetry Platform
EA-0020	IS-020	AQELYN AI Decision Intelligence Engine

The fixed repository structure remains:

```
AQELYN/  
■■■ archive/  
■■■ blueprint/  
■■■ docs/  
■■■ src/  
■■■ tests/  
■■■ tools/  
■■■ build/  
■■■ releases/  
■■■ scripts/  
■■■ assets/  
■■■ examples/  
■■■ plugins/  
■■■ sdk/  
■■■ api/  
■■■ README.md
```

The engineering rule remains:

```
Finish Implementation Specification  
↓  
Generate Engineering Archive  
↓  
Continue
```

No Engineering Archive may be skipped.

2. IS-020 Specification Identity

```
Specification ID: IS-020  
Name: AQELYN AI Decision Intelligence Engine  
Engineering Archive Target: EA-0020  
Project: AQELYN  
System Type: Cyber Security Operating Environment  
Status: Complete  
Predecessor: IS-019 - AQELYN Security Data Lake & Telemetry Platform
```

3. Purpose

The AQELYN AI Decision Intelligence Engine provides explainable, policy-governed artificial intelligence capabilities for decision support across the AQELYN platform.

It assists analysts, investigators, responders, compliance officers, and mission owners by transforming telemetry, evidence, knowledge graph relationships, historical incidents, and operational context into explainable recommendations.

The engine supports human decision-making while ensuring that final authority remains governed by AQELYN policies and authorized operators.

It answers:

- What is the most likely explanation?
- What action is recommended?
- What evidence supports the recommendation?
- What risks are associated?
- What historical incidents are similar?
- What confidence level exists?
- Can the recommendation be explained?
- Can the recommendation be audited?

4. Mission

The engine shall provide:

- Explainable AI
- Decision intelligence
- Recommendation engine
- Confidence scoring
- Risk-aware reasoning
- Mission-aware reasoning
- Evidence-aware reasoning
- Policy-aware recommendations
- Historical similarity analysis
- Analyst assistance
- Executive decision support
- Continuous learning

5. Scope

5.1 In Scope

- Decision recommendations
- Evidence reasoning
- Threat reasoning
- Risk reasoning
- Mission reasoning
- Analyst assistance
- Incident prioritization
- Response recommendations
- Investigation guidance
- Historical pattern analysis
- Knowledge graph reasoning
- Confidence estimation

5.2 Out of Scope

- Autonomous offensive actions
- Unapproved destructive actions
- Operating system AI assistants
- Business forecasting
- Financial prediction
- Consumer AI services

6. Dependencies

IS-020 depends on:

- IS-001 AQELYN Kernel
- IS-002 Universal Object Model
- IS-003 AQELYN Event Bus
- IS-004 AQELYN Evidence Engine
- IS-005 AQELYN Knowledge Graph
- IS-006 AQELYN Trust Engine
- IS-007 AQELYN Mission Engine
- IS-008 AQELYN Workflow Engine
- IS-009 AQELYN Policy Engine
- IS-010 AQELYN Compliance & Governance Engine
- IS-011 Identity & Access Governance Engine
- IS-012 Asset & Configuration Governance Engine
- IS-013 Risk Intelligence Engine
- IS-014 Threat Intelligence Fusion Engine
- IS-015 Security Operations Engine
- IS-016 Digital Forensics Engine
- IS-017 Threat Detection & Analytics Engine
- IS-018 Automated Response & Orchestration Engine
- IS-019 Security Data Lake & Telemetry Platform

7. High-Level Architecture

AQELYN AI Decision Intelligence Engine

-
- Recommendation Engine
- Reasoning Engine
- Confidence Engine
- Explainability Engine
- Similarity Engine
- Knowledge Graph Connector
- Evidence Connector
- Policy Connector
- Mission Connector
- Risk Connector
- Learning Engine
- Analytics Engine
- Event Publisher

8. Functional Requirements

FR-020-001 - Decision Recommendations

The engine shall generate recommendations for:

- Incident response
- Threat investigation
- Containment strategy
- Recovery planning
- Mission protection
- Compliance actions

FR-020-002 - Explainable AI

Every recommendation shall include:

- Supporting evidence
- Confidence score
- Reasoning summary
- Policy references
- Risk considerations
- Mission considerations

FR-020-003 - Similarity Analysis

Support comparison against:

- Historical incidents
- Threat campaigns
- Known attack patterns
- Previous investigations
- Recovery outcomes
- Response effectiveness

FR-020-004 - Confidence Assessment

Calculate:

- Evidence confidence
- Recommendation confidence
- Prediction confidence
- Risk confidence
- Trust score

FR-020-005 - Continuous Learning

Support:

- Feedback collection
- Model evaluation
- Recommendation refinement
- Knowledge updates
- Performance measurement

FR-020-006 - Decision Governance

Support:

- Human approval
- Policy enforcement
- Auditability
- Recommendation traceability
- Decision history
- Version control

FR-020-007 - Event Publication

Publish standardized events:

- recommendation.generated
- recommendation.accepted
- recommendation.rejected
- confidence.updated
- learning.completed
- model.evaluated

9. Non-Functional Requirements

The engine shall provide:

- Explainability
- Low-latency inference
- High availability
- Scalable reasoning
- Auditability
- Repository stability
- Backward compatibility
- Continuous operation

10. Core Decision Lifecycle

```

Evidence Received
↓
Context Collection
↓
Reasoning
↓
Recommendation
↓
Confidence Assessment
↓
Human Review
↓
Decision
↓
Learning

```

11. Internal Component Architecture

The AQELYN AI Decision Intelligence Engine is implemented as a modular AI decision-support platform integrated with the AQELYN Kernel, Knowledge Graph, Evidence Engine, Policy Engine, Risk Intelligence Engine, and Security Data Lake.

```

AQELYN AI Decision Intelligence Engine
■
■■■ Recommendation Engine
■■■ Reasoning Engine
■■■ Confidence Engine
■■■ Explainability Engine
■■■ Similarity Engine
■■■ Learning Engine
■■■ Analytics Engine
■■■ Knowledge Graph Connector
■■■ Evidence Connector
■■■ Policy Connector
■■■ Mission Connector
■■■ Risk Connector
■■■ Event Publisher

```

12. Component Specifications

12.1 Recommendation Engine

Generates explainable decision recommendations.

Capabilities:

```

Incident recommendations
Response recommendations
Recovery recommendations
Investigation guidance
Mission recommendations

```

12.2 Reasoning Engine

Performs multi-source reasoning.

Functions:

```

Evidence reasoning
Knowledge Graph reasoning

```

- Threat reasoning
- Mission reasoning
- Risk reasoning

12.3 Confidence Engine

Calculates confidence values.

Supports:

- Evidence confidence
- Recommendation confidence
- Trust score
- Prediction confidence
- Decision certainty

12.4 Explainability Engine

Provides transparent AI explanations.

Produces:

- Reasoning summary
- Evidence references
- Confidence explanation
- Policy references
- Decision trace

12.5 Similarity Engine

Compares against historical knowledge.

Supports:

- Incident similarity
- Threat similarity
- Response similarity
- Investigation similarity
- Recovery similarity

12.6 Learning Engine

Improves recommendations over time.

Supports:

- Feedback collection
- Performance evaluation
- Knowledge refinement
- Model assessment
- Recommendation optimization

12.7 Analytics Engine

Provides AI operational metrics.

Calculates:

- Recommendation accuracy
- Acceptance rate
- Confidence trends
- Model performance
- Operational statistics

13. Universal Object Model Extensions

13.1 Recommendation

```
Recommendation:  
  recommendation_id  
  confidence  
  explanation  
  status
```

13.2 DecisionRecord

```
DecisionRecord:  
  decision_id  
  approver  
  outcome  
  timestamp
```

13.3 ConfidenceScore

```
ConfidenceScore:  
  score_id  
  value  
  evidence_reference  
  rationale
```

13.4 LearningRecord

```
LearningRecord:  
  learning_id  
  feedback  
  effectiveness  
  model_version
```

14. Knowledge Graph Integration

Relationships:

```
Evidence  
↓  
supports  
↓  
Recommendation  
  
Recommendation  
↓  
requires  
↓  
Decision  
  
Decision  
↓  
updates  
↓  
LearningRecord  
  
LearningRecord  
↓  
improves  
↓  
Recommendation
```

15. Event Bus Integration

15.1 Recommendation Events

```
recommendation.generated  
recommendation.updated  
recommendation.accepted  
recommendation.rejected
```

15.2 Confidence Events

```
confidence.calculated  
confidence.updated
```

15.3 Learning Events

```
learning.started  
learning.completed  
model.evaluated
```

16. Evidence Engine Integration

Consumes:

```
Evidence metadata  
Artifact references  
Integrity scores  
Trust assessments
```

17. Knowledge Graph Integration Details

Provides:

```
Entity relationships  
Historical context  
Threat relationships  
Mission context  
Investigation history
```

18. Risk Intelligence Integration

Consumes:

```
Risk scores  
Business impact  
Mission impact  
Risk trends  
Threat prioritization
```

19. Security Data Lake Integration

Consumes:

```
Historical telemetry  
Operational metrics  
Detection history  
Response history  
Analytical datasets
```

20. Compliance Integration

Supports:

- Decision auditing
- Recommendation history
- Policy validation
- Regulatory reporting
- Traceability

21. Public APIs

21.1 Recommendation API

- GET /recommendations
- POST /recommendations
- GET /recommendations/{id}

21.2 Decision API

- GET /decisions
- POST /decisions

21.3 Learning API

- GET /learning
- POST /learning

21.4 Analytics API

- GET /analytics
- GET /confidence

22. Repository Impact

Implementation shall use the approved repository structure.

```
AQELYN/  
■■■ src/  
■ ■■■ ai_decision_engine/  
■■■ tests/  
■ ■■■ ai_decision_engine/  
■■■ docs/  
■ ■■■ ai_decision_engine/  
■■■ api/  
■ ■■■ ai_decision_engine/  
■■■ archive/
```

No top-level repository modifications are permitted.

23. Security Architecture

The AQELYN AI Decision Intelligence Engine is the trusted decision-support subsystem responsible for producing explainable, policy-governed AI recommendations across the AQELYN platform.

Every recommendation shall be:

- Explainable
- Evidence-backed
- Policy-governed
- Risk-aware
- Mission-aware
- Fully auditable
- Traceable
- Human-reviewable

23.1 Security Principles

- Zero Trust
- Human-in-the-Loop
- Explainable AI
- Least Privilege
- Defense in Depth
- Continuous Validation
- Policy Enforcement
- Secure by Design

23.2 Authorization Model

Supported operational roles:

- SOC Analyst
- Threat Hunter
- Incident Commander
- Mission Owner
- Compliance Officer
- Security Administrator
- AI Administrator
- Automation Service

All AI-generated recommendations shall be governed by the AQELYN Policy Engine.

23.3 Recommendation Integrity

Recommendation records shall maintain:

- Unique recommendation identifier
- Model or reasoning version
- Evidence references
- Confidence score
- Explanation
- Policy references
- Decision state
- Audit trail

Recommendation history shall be append-only.

23.4 Explainability Integrity

Explanations shall maintain:

- Evidence references
- Reasoning summary
- Confidence rationale
- Policy rationale
- Risk rationale
- Mission rationale
- Decision trace

No recommendation shall be considered operationally valid without an explanation record.

24. Decision Lifecycle

24.1 Recommendation Lifecycle

```
Evidence Collected
↓
Reasoning
↓
Recommendation Generated
↓
Confidence Calculated
↓
Human Review
↓
Accepted / Rejected
↓
Recorded
```

24.2 Learning Lifecycle

```
Feedback Received
↓
Evaluation
↓
Model Assessment
↓
Knowledge Update
↓
Learning Complete
```

24.3 Decision Audit Lifecycle

```
Decision Recorded
↓
Evidence Linked
↓
Policy Verified
↓
Audit Stored
```

25. Continuous AI Operations

The engine continuously evaluates:

```
Recommendation quality
Confidence accuracy
Model performance
Feedback trends
Historical effectiveness
Policy compliance
Mission impact
Operational risk
```

26. Performance Requirements

The engine shall support:

```
Low-latency inference
Enterprise-scale reasoning
Concurrent recommendations
High-speed similarity analysis
Continuous learning
Continuous availability
```

27. Scalability Requirements

The engine shall scale to support:

- Global deployments
- Large enterprise environments
- Millions of historical recommendations
- Distributed inference services
- Hybrid cloud deployments
- Multi-region operations

28. Audit Requirements

Every AI operation shall generate immutable audit records.

Audit events include:

- Recommendation generated
- Recommendation accepted
- Recommendation rejected
- Confidence calculated
- Learning completed
- Model evaluated
- Policy validation

29. Failure Handling

29.1 Recommendation Failure

- Recommendation rejected
- Failure recorded
- Analyst notified

29.2 Confidence Failure

- Confidence recalculated
- Audit generated
- Fallback explanation provided

29.3 Learning Failure

- Learning suspended
- Model preserved
- Administrator notified

29.4 Policy Failure

- Recommendation blocked
- Policy violation recorded
- Human review required

30. Testing Strategy

30.1 Unit Testing

Validate:

```
Recommendation Engine
Reasoning Engine
Confidence Engine
Explainability Engine
Learning Engine
Analytics Engine
```

30.2 Integration Testing

Verify interaction with:

```
Kernel
Knowledge Graph
Evidence Engine
Policy Engine
Risk Intelligence Engine
Security Data Lake
Compliance Engine
```

30.3 System Testing

Validate:

```
Recommendation generation
Explainability
Similarity analysis
Confidence calculation
Learning workflows
Decision auditing
```

30.4 Security Testing

Verify:

```
Authorization
Policy enforcement
Explainability
Audit logging
Recommendation integrity
```

30.5 Regression Testing

Verify IS-001 through IS-019 remain unaffected.

31. Acceptance Criteria

IS-020 is complete when:

```
Recommendation Engine implemented
Reasoning Engine implemented
Confidence Engine implemented
Explainability Engine implemented
Learning Engine implemented
Repository unchanged
Testing documented
```

32. Repository Validation

Repository structure remains unchanged.

```
AQELYN/
■■■ src/ai_decision_engine/
```

```
■■■ tests/ai_decision_engine/  
■■■ docs/ai_decision_engine/  
■■■ api/ai_decision_engine/  
■■■ archive/
```

No top-level repository modifications are permitted.

33. Engineering Summary

IS-020 introduces the AQELYN AI Decision Intelligence Engine, providing explainable AI recommendations, confidence scoring, historical reasoning, similarity analysis, policy-governed decision support, and continuous learning.

Major capabilities include:

```
Decision Intelligence  
Explainable AI  
Recommendation Engine  
Confidence Scoring  
Historical Reasoning  
Similarity Analysis  
Learning Engine  
Decision Auditing  
Policy-aware AI  
Mission-aware AI  
Risk-aware AI
```

The engine integrates with all previously completed AQELYN engines while preserving repository stability, modularity, and backward compatibility.

34. Specification Status

```
Specification ID : IS-020  
Title : AQELYN AI Decision Intelligence Engine  
Status : COMPLETE  
Engineering Archive : READY FOR GENERATION  
Next Artifact : EA-0020
```

Engineering workflow status:

```
EA-0001 COMPLETE  
EA-0002 COMPLETE  
EA-0003 COMPLETE  
EA-0004 COMPLETE  
EA-0005 COMPLETE  
EA-0006 COMPLETE  
EA-0007 COMPLETE  
EA-0008 COMPLETE  
EA-0009 COMPLETE  
EA-0010 COMPLETE  
EA-0011 COMPLETE  
EA-0012 COMPLETE  
EA-0013 COMPLETE  
EA-0014 COMPLETE  
EA-0015 COMPLETE  
EA-0016 COMPLETE  
EA-0017 COMPLETE  
EA-0018 COMPLETE  
EA-0019 COMPLETE  
IS-020 COMPLETE  
EA-0020 READY FOR GENERATION
```

35. EA-0020 Engineering Objective

The objective of IS-020 was to introduce a dedicated AI Decision Intelligence Engine that enables AQELYN to produce explainable, evidence-backed, policy-governed recommendations for security operations, incident response, mission protection, risk management, compliance actions, and investigation support.

The engine extends AQELYN from telemetry and automation into governed AI-assisted decision intelligence.

36. EA-0020 Engineering Summary

The implementation specification defines a modular subsystem responsible for:

- Recommendation generation
- Multi-source reasoning
- Confidence scoring
- Explainability
- Similarity analysis
- Continuous learning
- Analytics
- Knowledge Graph integration
- Evidence integration
- Policy integration
- Mission integration
- Risk integration
- Security Data Lake integration
- Event publishing

The engine integrates with all previously completed AQELYN engines while preserving architectural modularity.

37. Major Engineering Decisions

37.1 Decision 1 - Dedicated AI Decision Intelligence Engine

AI decision support is implemented as a standalone engine rather than embedded in SOC, Risk Intelligence, or Automated Response.

Rationale:

- Clear separation of AI-assisted reasoning from operational execution.
- Independent lifecycle and governance.
- Better support for explainability, confidence scoring, and learning.
- Improved auditability of recommendations and decision outcomes.

37.2 Decision 2 - Human-in-the-Loop AI

AI recommendations support decisions but do not bypass authorized human or policy-governed control.

Benefits:

- Final authority remains governed.
- High-impact actions require review where policy requires it.
- Recommendation acceptance and rejection are auditable.
- Mission and risk owners can remain part of decision workflows.

37.3 Decision 3 - Explainability as Mandatory

No recommendation is operationally valid without an explanation.

Benefits:

Analysts can understand reasoning.
 Executives can trust recommendations.
 Compliance can audit decisions.
 False or weak recommendations can be reviewed and improved.

37.4 Decision 4 - Evidence-Backed Recommendations

Recommendations must reference evidence, confidence, policy, mission, and risk context.

Benefits:

Recommendations become traceable.
 Decision records are defensible.
 Risk and mission implications remain visible.
 Compliance reporting can use recommendation history.

37.5 Decision 5 - Event-Driven AI Decision Lifecycle

Recommendation, confidence, decision, learning, and model evaluation events are published through the AQELYN Event Bus.

Examples include:

```
recommendation.generated
recommendation.updated
recommendation.accepted
recommendation.rejected
confidence.calculated
confidence.updated
learning.started
learning.completed
model.evaluated
```

This maintains loose coupling between AQELYN engines.

37.6 Decision 6 - Universal Object Model Extension

New domain objects introduced include:

```
Recommendation
DecisionRecord
ConfidenceScore
LearningRecord
```

These extend the Universal Object Model without modifying existing object definitions.

38. Architectural Integration Summary

| Engine | Integration |

|---|---|

| IS-001 Kernel | Runtime lifecycle and service registration |

| IS-002 Universal Object Model | Recommendation, decision, confidence, learning objects |

| IS-003 Event Bus | Recommendation, decision, confidence, learning events |

| IS-004 Evidence Engine | Evidence metadata, artifact references, trust assessments |

| IS-005 Knowledge Graph | Entity relationships, historical context, investigation history |

| IS-006 Trust Engine | Trust score and evidence confidence |

| IS-007 Mission Engine | Mission impact and mission-aware reasoning |

| IS-008 Workflow Engine | Decision review, approval, feedback, learning workflows |

IS-009 Policy Engine	Recommendation governance and policy validation
IS-010 Compliance Engine	Decision auditing, regulatory reporting, traceability
IS-013 Risk Intelligence Engine	Risk scores, impact, trends, prioritization
IS-014 Threat Intelligence Engine	Threat context, campaigns, actor profiles, indicators
IS-015 SOC Engine	Analyst assistance, incident support, investigation guidance
IS-016 Digital Forensics Engine	Forensic reports, evidence artifacts, timelines
IS-017 Threat Detection Engine	Detections, anomalies, behavior profiles, predictions
IS-018 Response Orchestration Engine	Response recommendations and playbook support
IS-019 Security Data Lake	Historical telemetry, metrics, response and detection history

No existing engine required redesign.

39. Repository Impact Summary

Repository structure remains unchanged.

Implementation is expected within existing project directories, including:

```
AQELYN/  
■■■ src/ai_decision_engine/  
■■■ tests/ai_decision_engine/  
■■■ api/ai_decision_engine/  
■■■ docs/ai_decision_engine/  
■■■ archive/
```

No top-level directories were added, removed, or renamed.

40. Security Impact Summary

The specification introduces AI-decision-specific security controls:

```
Policy-governed recommendations  
Human-in-the-loop review  
Mandatory explainability  
Evidence-backed reasoning  
Confidence scoring  
Recommendation audit trail  
Decision history  
Learning governance  
Model evaluation records  
Role-authorized AI administration
```

No reduction in the security posture of existing components was identified.

41. Capabilities Added

The engine enables AQELYN to support:

```
Explainable AI  
Decision intelligence  
Recommendation generation  
Confidence scoring  
Risk-aware reasoning  
Mission-aware reasoning  
Evidence-aware reasoning  
Policy-aware recommendations  
Historical similarity analysis
```

Analyst assistance
Executive decision support
Continuous learning
Recommendation auditing

42. Risks Identified

| Risk | Mitigation |

|---|---|

| Unexplainable recommendation | Mandatory Explainability Engine |

| Over-trusting AI output | Human-in-the-loop governance and policy enforcement |

| Weak evidence basis | Evidence references and confidence scoring |

| Model drift | Learning evaluation and model assessment |

| Policy-violating recommendations | Policy Connector and recommendation blocking |

| Recommendation bias | Audit, feedback collection, evaluation metrics |

| Poor historical similarity | Similarity confidence and evidence context |

| Unauthorized AI administration | Role-based authorization and audit |

No critical architectural risks were identified that require redesign.

43. Verification Summary

The specification defines verification for:

Unit testing
Integration testing
System testing
Security testing
Regression testing

Acceptance criteria cover recommendation engine, reasoning engine, confidence engine, explainability engine, learning engine, repository validation, and testing documentation.

44. Engineering Principles Confirmed

The implementation complies with established AQELYN principles:

Modular architecture
Event-driven communication
Immutable evidence references
Traceability
Explainability
Security by design
Repository stability
Backward compatibility
Governance-before-archive discipline

45. Dependencies

Required:

EA-0001 through EA-0019
IS-001 through IS-019

Enables:

IS-021 and subsequent predictive analytics, forecasting, executive intelligence, and AI-assisted platform components

46. Completion Record

Engineering Archive : EA-0020
Implementation Specification : IS-020
Title : AQELYN AI Decision Intelligence Engine
Engineering Status : COMPLETE
Repository Status : UNCHANGED
Architecture Status : EXTENDED
Backward Compatibility : MAINTAINED
Engineering Rule :
IS Completed
↓
EA Generated
↓
Continue

47. Archive Index Update

EA-0001 IS-001 AQELYN Kernel
EA-0002 IS-002 Universal Object Model
EA-0003 IS-003 AQELYN Event Bus
EA-0004 IS-004 AQELYN Evidence Engine
EA-0005 IS-005 AQELYN Knowledge Graph
EA-0006 IS-006 AQELYN Trust Engine
EA-0007 IS-007 AQELYN Mission Engine
EA-0008 IS-008 AQELYN Workflow Engine
EA-0009 IS-009 AQELYN Policy Engine
EA-0010 IS-010 AQELYN Compliance & Governance Engine
EA-0011 IS-011 AQELYN Identity & Access Governance Engine
EA-0012 IS-012 AQELYN Asset & Configuration Governance Engine
EA-0013 IS-013 AQELYN Risk Intelligence Engine
EA-0014 IS-014 AQELYN Threat Intelligence Fusion Engine
EA-0015 IS-015 AQELYN Security Operations (SOC) Engine
EA-0016 IS-016 AQELYN Digital Forensics Engine
EA-0017 IS-017 AQELYN Threat Detection & Analytics Engine
EA-0018 IS-018 AQELYN Automated Response & Orchestration Engine
EA-0019 IS-019 AQELYN Security Data Lake & Telemetry Platform
EA-0020 IS-020 AQELYN AI Decision Intelligence Engine

48. Engineering Phase Status

Completed Engineering Archives : EA-0001 through EA-0020

Current Status:
EA-0020 COMPLETE

Next Implementation Specification:
IS-021 - AQELYN Predictive Analytics & Forecasting Engine

EA-0020 is completed and archived. The engineering workflow is consistent with the project rule:

Implementation Specification -> Engineering Archive -> Continue

From this point onward, the next engineering artifact is IS-021.

49. Engineering Archive Publication Standard

EA-0020 follows the AQELYN Engineering Archive Publication Standard.

Required package structure:

```
EA-xxxx/  
■  
■ ■ ■ diagrams/  
■ ■ ■ ■ Architecture.svg  
■ ■ ■ ■ Component.svg  
■ ■ ■ ■ Workflow.svg  
■ ■ ■ ■ EventFlow.svg  
■ ■ ■ ■ Integration.svg  
■  
■ ■ ■ examples/  
■ ■ ■ ■ example_*.md  
■  
■ ■ ■ HTML/  
■ ■ ■ ■ index.html  
■ ■ ■ ■ styles.css  
■ ■ ■ ■ assets/  
■  
■ ■ ■ images/  
■  
■ ■ ■ journal/  
■ ■ ■ ■ Engineering_Journal.md  
■  
■ ■ ■ manifest/  
■ ■ ■ ■ manifest.json  
■  
■ ■ ■ MD/  
■ ■ ■ ■ EA-xxxx.md  
■  
■ ■ ■ PDF/  
■ ■ ■ ■ EA-xxxx.pdf  
■  
■ ■ ■ requirements/  
■ ■ ■ ■ Requirements_Matrix.md  
■  
■ ■ ■ traceability/  
■ ■ ■ ■ Traceability_Matrix.md  
■  
■ ■ ■ README.md
```

The master Markdown is the source of truth. The PDF and HTML are generated from the same master Markdown and must not omit sections.

50. Requirements Matrix

Requirement ID	Requirement	Evidence in Archive	Status
FR-020-001	Generate decision recommendations	Sections 8, 12, 24	Complete
FR-020-002	Provide explainable AI	Sections 8, 12, 23	Complete
FR-020-003	Support similarity analysis	Sections 8, 12	Complete
FR-020-004	Calculate confidence	Sections 8, 12, 23	Complete
FR-020-005	Support continuous learning	Sections 8, 12, 24	Complete
FR-020-006	Support decision governance	Sections 8, 23, 24	Complete
FR-020-007	Publish AI decision events	Sections 8, 15, 37	Complete
NFR-020-001	Explainability	Sections 9, 23, 37	Complete
NFR-020-002	Low-latency inference	Sections 9, 26	Complete
NFR-020-003	High availability	Sections 9, 26	Complete
NFR-020-004	Scalable reasoning	Sections 9, 27	Complete

| NFR-020-005 | Auditability | Sections 9, 28, 40 | Complete |

| NFR-020-006 | Repository stability | Sections 22, 32, 39 | Complete |

51. Traceability Matrix

Source	Target	Relationship
IS-020 Purpose	EA-0020 Objective	Defines why the engine exists
Recommendation Engine	FR-020-001	Generates recommendations
Explainability Engine	FR-020-002	Produces explanation records
Similarity Engine	FR-020-003	Compares historical incidents and patterns
Confidence Engine	FR-020-004	Calculates confidence
Learning Engine	FR-020-005	Supports feedback and refinement
Policy Connector	FR-020-006	Enforces decision governance
Event Publisher	FR-020-007	Publishes recommendation, confidence, and learning events
Evidence Engine Integration	Evidence-backed reasoning	Supplies evidence metadata and trust
Knowledge Graph Integration	Reasoning context	Supplies entity relationships and history
Risk Intelligence Integration	Risk-aware reasoning	Supplies risk scores and trends
Security Data Lake Integration	Historical context	Supplies telemetry and operational history
Compliance Integration	Decision auditing	Supports audit and regulatory reporting
Repository Validation	Repository Standard	Confirms no top-level redesign

52. Engineering Journal

Journal Entry - EA-0020

EA-0020 was created to archive completion of IS-020 - AQELYN AI Decision Intelligence Engine.

The archive records the expansion of AQELYN into governed AI-assisted decision intelligence. IS-020 defines the structure needed to generate recommendations, reason across evidence and knowledge graph context, calculate confidence, explain recommendations, compare against historical incidents, collect feedback, support learning, audit decisions, and integrate with policy, mission, risk, compliance, and telemetry services.

The engineering design preserves the fixed AQELYN repository structure and maintains backward compatibility with previously completed engines.

Lessons Learned

AI decision intelligence must be modeled separately from automated response and SOC operations. AI can recommend, explain, and learn, but policy and human governance determine whether recommendations become operational actions.

Governance Note

EA-0020 follows the master-document publication workflow. The Markdown file is the authoritative source, and PDF/HTML representations are generated from the same content.

53. Examples

53.1 Example Recommendation

```
recommendation_id: REC-0001
confidence: 0.87
status: pending_review
explanation: High-confidence behavioral anomaly with matching threat intelligence and mission impact.
evidence:
- evidence://detection-1001
- evidence://risk-score-asset-0002
policy_references:
- policy://response-human-review-required
```

53.2 Example Decision Record

```
decision_id: DEC-1001
recommendation_id: REC-0001
approver: incident_commander_01
outcome: accepted
timestamp: 2026-07-07T12:10:00Z
reason: Evidence supports containment recommendation.
```

53.3 Example Confidence Score

```
score_id: CONF-2001
value: 0.87
evidence_reference: evidence://detection-1001
rationale: Evidence quality high, threat confidence high, historical match moderate.
```

53.4 Example AI Decision Event

```
{
  "event_type": "recommendation.generated",
  "recommendation_id": "REC-0001",
  "confidence": 0.87,
  "status": "pending_review",
  "source_engine": "aqelyn_ai_decision_intelligence_engine"
}
```

54. Manifest Summary

Archive contents include:

```
README.md
MD/EA-0020.md
PDF/EA-0020.pdf
HTML/index.html
HTML/styles.css
manifest/manifest.json
requirements/Requirements_Matrix.md
traceability/Traceability_Matrix.md
journal/Engineering_Journal.md
diagrams/Architecture.svg
diagrams/Component.svg
diagrams/Workflow.svg
diagrams/EventFlow.svg
diagrams/Integration.svg
examples/example_ai_decision.md
```

55. Final Archive Statement

EA-0020 is the Engineering Archive for IS-020 - AQELYN AI Decision Intelligence Engine.

It records the completed specification, the architectural decisions, the integration model, the repository impact, the risk posture, verification requirements, acceptance criteria, archive index update, and the engineering publication standard.

EA-0020 COMPLETE
IS-020 COMPLETE
NEXT: IS-021